

IZAK OLTMAN

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RESEARCH INTERESTS

Semiclassical analysis and random perturbations of non-self-adjoint operators.

EMPLOYMENT

Northwestern University

Fall 2024 - Present

NSF RTG Fellow and Boas Assistant Professor

Postdoc Supervisor: Jared Wunsch

EDUCATION

University of California, Berkeley

Fall 2019 - Spring 2024

Ph.D. in Mathematics

Advisor: Maciej Zworski

University of Wisconsin, Madison

Fall 2015 - Spring 2019

Bachelor of Arts – College of Letters and Science

Major in Mathematics with Honors

Certificate in Physics

Budapest Semesters in Mathematics

Spring 2018

Budapest, Hungary

AWARDS

NSF Graduate Research Fellowship

2021 - 2024

Dowling Scholarship - UW Madison Math Department

2018

Dean's List - UW Madison

2015 - 2019

Vilas Scholarship - UW Madison

2015 - 2018

AMETEK Scholarship

2015 - 2019

Green House Summer Research Scholarship - UW Madison

Summer 2016

PUBLICATIONS AND PREPRINTS

10. *The spectrum of the Scottish flag operator* with Frédéric Klopp (*in preparation*)
9. *Spectra of Non-Self-Adjoint Almost Mathieu Matrices and the Scottish Flag Operator* with Simon Becker ([arXiv:2608.30191](https://arxiv.org/abs/2608.30191))
8. *The asymptotic structure of forward scattering* with Nicholas Lohr, Ethan Sussman, and Yuzhou Joey Zou ([arXiv:2608.27815](https://arxiv.org/abs/2608.27815))
7. *A Poisson Formula for the Wave Propagator on Schwarzschild-de Sitter Backgrounds* with Benjamin Pineau ([arXiv:2512.16054](https://arxiv.org/abs/2512.16054)) – to appear in **Communications in Mathematical Physics**
6. *Spectral instability of random Fredholm operators* with Simon Becker and Martin Vogel – in **Journal of Spectral Theory**, 16.1(2026), 335-389 ([arXiv:2502.10222](https://arxiv.org/abs/2502.10222))
5. *Absence of small magic angles for disordered tunneling potentials in twisted bilayer graphene* with Simon Becker and Martin Vogel ([arXiv:2402.12799](https://arxiv.org/abs/2402.12799)) – to appear in the **AMS Contemporary Mathematics** volume in memory of Steve Zelditch
4. *Magic angle (in)stability and mobility edges in disordered Chern insulators* with Simon Becker and Martin Vogel – in **Journal of Physics A: Mathematical and Theoretical**, 58.36(2025) ([arXiv:2309.02701](https://arxiv.org/abs/2309.02701))

3. *Particle trajectories for quantum maps* with Yonah Borns-Weil ([arXiv:2210.03224](https://arxiv.org/abs/2210.03224)) – in **Annales Henri Poincaré**, 25.8(2024), 3699-3738
2. *A probabilistic Weyl-law for perturbed Berezin–Toeplitz operators* – in **Journal of Spectral Theory**, 13.2(2023), 727-754 ([arXiv:2207.09599](https://arxiv.org/abs/2207.09599))
1. *An exotic calculus of Berezin–Toeplitz operators* – in **Mathematika**, 2.2(2025) ([arXiv:2207.09596](https://arxiv.org/abs/2207.09596))

TALKS/PRESENTATIONS

- *AMS Special Session on Microlocal Analysis and Mathematical Physics*. March 2026. Boston College.
- *Inverse Problems, Spectral, and Scattering Theory Seminar*. March 2026. Purdue University.
- *Analysis and Applied Mathematics Seminar*. February 2026. Bocconi University.
- *Spectral Theory Seminar*. January 2026. UC Berkeley.
- *RTG Dynamics REU Colloquium*. July 2025. Northwestern University.
- *Analysis Seminar*. February 2025. Northwestern University.
- *Probability and Analysis Seminar*. February 2025. University of Michigan, Ann Arbor.
- *Spectral Theory Seminar*. January 2025. UC Berkeley.
- *Joint Mathematics Meeting*. January 2025. Seattle, Washington.
- *International workshop on 2D and moiré materials*. July 2024. Roscoff, France.
- *Harmonic Analysis and Differential Equations Seminar*. February 2024. UC Berkeley.
- *Analysis Seminar*. November 2023. University of Strasbourg.
- *Spectral Theory Seminar*. November 2023. UC Berkeley.
- *Microlocal and Probabilistic Methods in Geometry and Dynamics Summer School (poster presentation)*. July 2023. Paris, France.
- *The 24th Midrasha Mathematicae Random Schrödinger Operators and Random Matrices (poster presentation)*. May 2023. IIAS Jerusalem.
- *Harmonic Analysis and Differential Equations Seminar*. April 2023. UC Berkeley.
- *PDE and Mathematical Physics Seminar*. November 2022. ETH Zurich.
- *Probability and Mathematical Physics Conference (poster presentation)*. July 2022. Helsinki, Finland.
- *Mathematical Challenges in Quantum Mechanics (poster presentation)*. June 2022. Como, Italy.
- *6th Great Lakes Mathematical Physics Conference*. June 2022. Michigan State University.
- *Harmonic Analysis and Differential Equations Seminar*. February 2022. UC Berkeley.

TEACHING

Instructor at Northwestern University

- Math 230-1: multivariable differential calculus (Spring 2026)
- Math 228-1: multivariable differential calculus for Engineering (Fall 2025)
- Math 230-2: multivariable integral calculus (Spring 2025)
- Math 228-1: multivariable differential calculus for Engineering (Fall 2024)

Graduate Student Instructor at UC Berkeley:

- Math 128a: numerical analysis (Spring 2021)

- Math 16a: calculus and analytic geometry (Fall 2020)
- Stat 134: concepts of probability (Summer 2020)
- Math 53: multivariable calculus (Spring 2020)
- Math 54: linear algebra and differential equations (Fall 2019)

CODING LANGUAGES

$\LaTeX$, MATLAB, Python

SERVICE

Co-organizer, AMS Special Session, 2027 Joint Mathematics Meetings (January 2027)

Organizer and mentor of an REU at Northwestern (2025); mentored Reyna Li and Ian Hollas on modeling quantum trajectories on Kähler manifolds

Officer of Berkeley's Math Graduate Student Association (2020 - 2023)

Private Math Tutor for Berkeley Undergraduates (2019 - 2024)

Organizer of UC Berkeley's Semi-classical Analysis Seminar (2022)

Organizer of UC Berkeley's Harmonic Analysis and Differential Equations Student Seminar (2023 - 2024)